VNIVERSIDAD D SALAMANCA

# Renal Physiology

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Welcome to **Renal Physiology**.

# Content

In this book you will find:

- Detailed analysis of hydroelectrolytic balance

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# HYDROELECTROLYTIC BALANCE

## 1.1 Properties of Water

- Strong polarity: Polar molecules attract ions and other polar compounds, allowing them to dissociate.

*This fact allows many types of molecules to dissolve in cells, enabling a great variety of chemical reactions and the transport of numerous substances.*

- High specific heat: Hydrogen bonds absorb heat when they break and release it when they form, minimizing temperature changes; that is, it can lose and acquire large amounts of heat with small changes in temperature.

*This allows body temperature to remain constant.*

- High heat of vaporization: For water to evaporate, many hydrogen bonds must be broken, which means that water requires a large absorption of heat to go from liquid to gas.

*This fact allows the evaporation of water through sweat to cool the body.*

- High cohesion: Hydrogen bonds hold water molecules together.

*Water acts as a lubricant to protect the body against injuries from friction or trauma.*

## 1.2 Fluid Compartments in the Body

Water is undoubtedly the most abundant component of the body, making up 45 to 75% of its weight. This large variation in water content is fundamentally dependent on the amount of adipose tissue, so the percentage of body weight represented by water will vary inversely with the body's fat content.

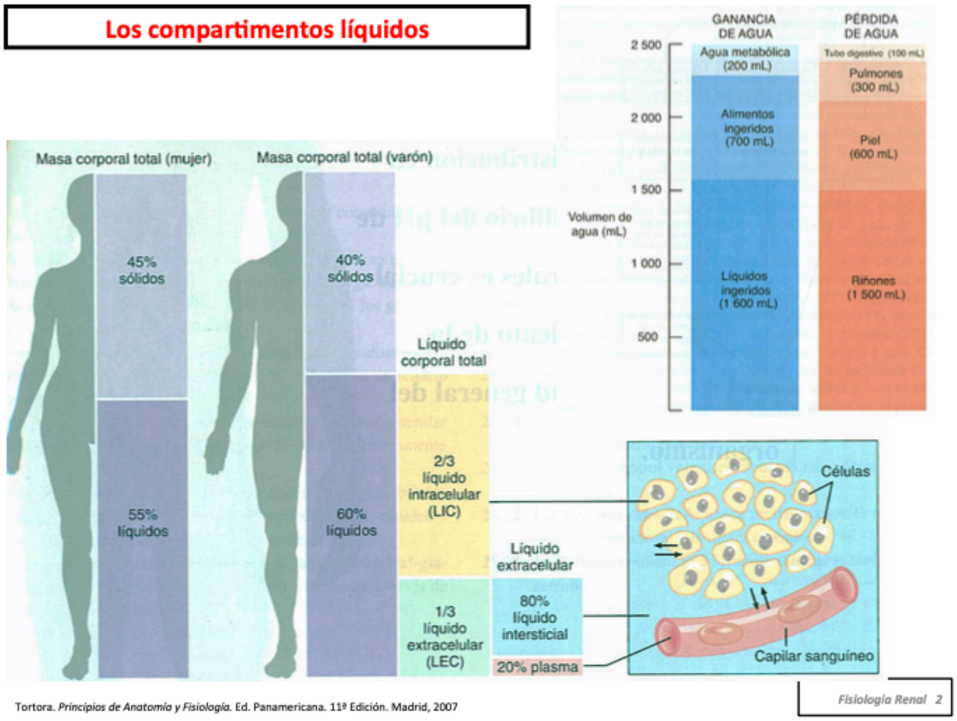
Total body water is around 60% of the weight in men and 55% in women, a percentage that will decrease as we age.

The cells of the body live in an "internal sea" of extracellular fluid (**ECF**) enclosed within the integuments, making up 1/3 of total body fluid. From this fluid, cells take oxygen and nutrients, and into it they eliminate their metabolic waste products.

This ECF is more dilute than modern oceans, but its composition is possibly similar to that of the primitive oceans where life is supposed to have originated.

The ECF is divided into two compartments:

- **Interstitial fluid**
- Circulating blood **Plasma**
- **Transcellular fluid**: fluid of the gastrointestinal, biliary, and urinary tracts, intraocular and cerebrospinal fluids, and the fluid of serous spaces such as pleural, peritoneal, and pericardial spaces.




The remaining 2/3 of body fluid is enclosed inside the cells and is called Intracellular Fluid (**ICF**).

**The barriers separating the different compartments are:**

1. **Plasma membranes**: they separate the intracellular from the interstitial fluid and are semi-permeable, highly selective membranes. Passive and active movements determine the different electrolyte concentrations on both sides.
2. **Capillary endothelium**.

## 1.3 Hydroelectrolytic Balance

Through the processes of filtration, reabsorption, diffusion, or osmosis, water and the solutes dissolved in it are stabilized on both sides of the barriers. Osmosis is the main mechanism for water movement, and therefore it is the concentration of solutes in the different fluids that determines the direction of water movement.

Compounds, depending on the types of chemical bonds, may or may not dissociate in water. Compounds that dissociate are called **electrolytes**, like sodium chloride, which dissociates into positive ions like sodium and negative ions like chloride. There are compounds that do not dissociate, like glucose, and they are known as **non-electrolytes**.

The **electrolytic composition** of the different compartments can be summarized in the following table:

| | Intracellular | Extracellular Intravascular (plasma) | Extracellular Interstitial |
|---|---|---|---|
| **Sodium** | 14-15 meq/L | 140-150 meq/L | 143 meq/L |
| **Potassium** | 140-150 meq/L | 4-5 meq/L | 4 meq/L |
| **Chloride** | 9 meq/L | 107-125 meq/L | 117 meq/L |
| **Bicarbonate** | 10 meq/L | 27 meq/L | 27 meq/L |
| **Proteins** | 74 meq/L | 16 meq/L | 2 meq/L |

The input and output of water in the body is in balance.

**Daily gain** is usually around 2,500 ml per day, coming from:

1. Metabolism, which provides about 200 ml of water per day.
2. Food, which provides about 700 ml per day.
3. Liquid intake, which provides about 1,600 ml per day.

**Water losses** occur through several mechanisms:

1. Digestive system: approximately 100 ml per day.
2. Respiration: accounts for about 300 ml per day of loss.
3. Transpiration through the skin: about 600 ml per day.
4. Renal: 1,500 ml per day.

To maintain the balance between water gain and loss, the kidney plays a crucial role through:

- Formation of **Dilute Urine** with which it excretes excess water.
- Formation of **Concentrated Urine** with which it excretes excess electrolytes.

## 1.4 Mechanisms of Water Regulation

Water balance is determined by mechanisms that can modify fluid elimination so that it matches intake, and by mechanisms that adjust intake to match elimination.

However, the mechanisms that control the passage of water between fluid compartments are the ones that act fastest, doing so at the expense of interstitial fluid to maintain plasma volume.

### 1.4.1 Regulation of Water Intake

The following elements are involved:

- Although the mechanisms adjusting fluid intake when elimination increases and vice versa are unknown, there is a group of neurons located in the roof of the third ventricle known as the **subfornical organ (SFO)** that act in fluid homeostasis.
- Supraoptic and paraventricular nuclei of the hypothalamus secreting **antidiuretic hormone (ADH)**.
- Osmoreceptors located in the brain.

These two nervous centers are interconnected, known as the **thirst center**, and are composed of **osmoreceptors**.

These mechanisms are stimulated by an increase in solute concentration and a decrease in saliva, which increases the sensation of thirst. Their mechanisms of action are summarized in the figure.

### 1.4.2 Regulation of Eliminated Volume

The control mechanisms act on the glomerular filtration rate and on the rate of water reabsorption in the tubules.

CHAPTER 2

# BIBLIOGRAPHY